Histology of Nervous System

Prof Eman Faruk

Learning Objectives

- List the cell types of the nervous system and their functions.
- Recognize the component parts of a neuron and their functions.
- Recognize the morphology of different neuronal types within a variety of CNS structures, e.g. cerebral cortex cerebellum, dorsal horn and ventral horn of spinal cord.
- Recognize the different types of glial cells, their structure, function and location.
- Recognize the cytological and architectural organization of the cerebrum, cerebellum, brain stem and spinal cord.

Outlines

- Overview—Nervous System
- Nervous cells
- Neurons and types
- Types of neuroglial

Overview—Nervous System

Organized into

- **Anatomically**
  - 1-Central nervous system CNS (brain and spinal cord)
  - 2-Peripheral nervous system(includes the nerves outside the CNS and their associated ganglia.
- **Functionally**

Overview—Nervous System

Organized into

Anatomically

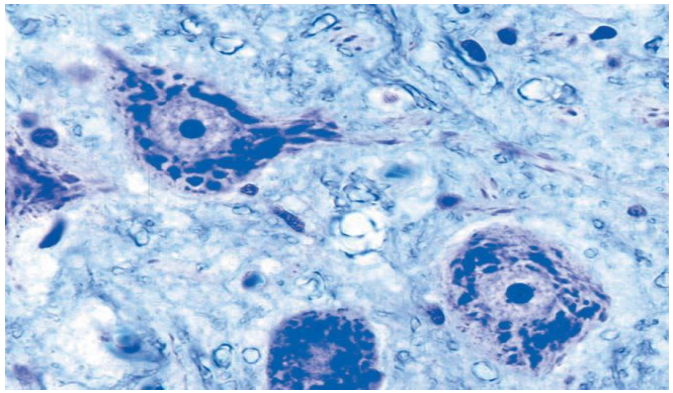
Functionally

- Sensory component, which transmits electrical impulses (signals) to the CNS
- Motor component, which transmits impulses from the CNS to various structures of the body

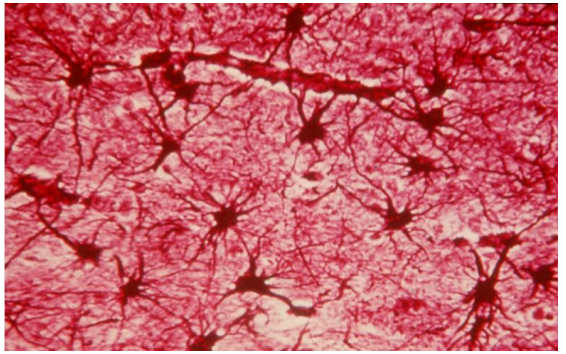
The motor component is further divided into the somatic and autonomic systems. It includes the cranial and spinal nerves and their associated ganglia

Cells of The Nervous System

- Neurons:
  - Excitability
  - Conductivity
  - Post-mitotic, fully differentiated cells
  - 10-50 times less numerous than glial cells

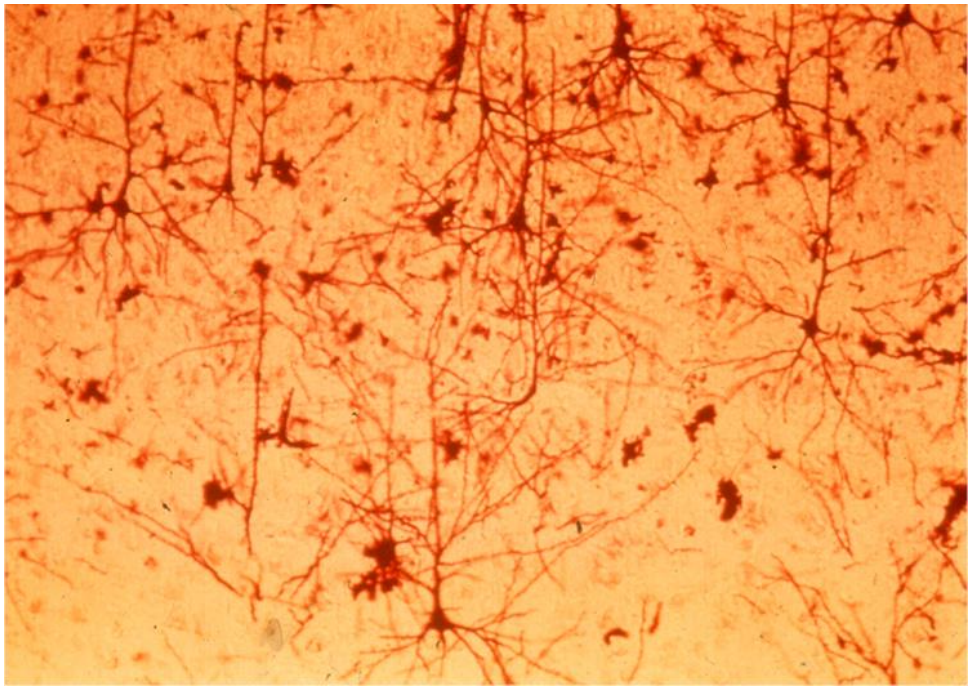


- Neuroglia:
  - The name for glial cells derives from Greek for glue
  - Providing structural and functional support for neuronal elements
  - Glial cells account for ~ 90% of all CNS cells



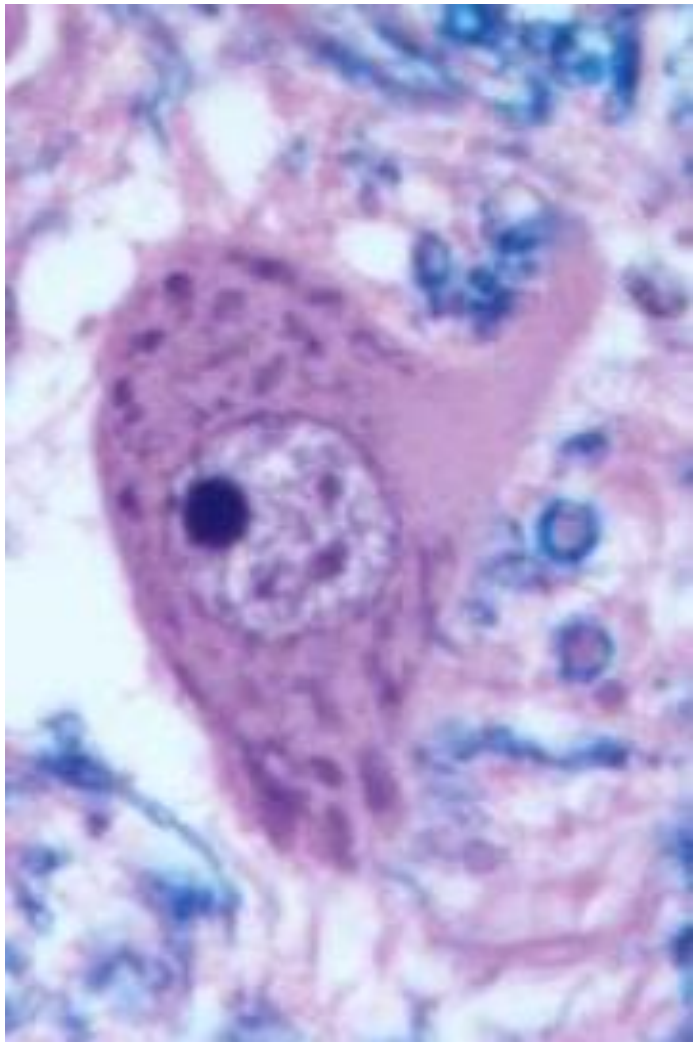
Neurons

- The neuron consists of:
  1. The cell body.
  2. Processes (single axon and one or more dendrites).

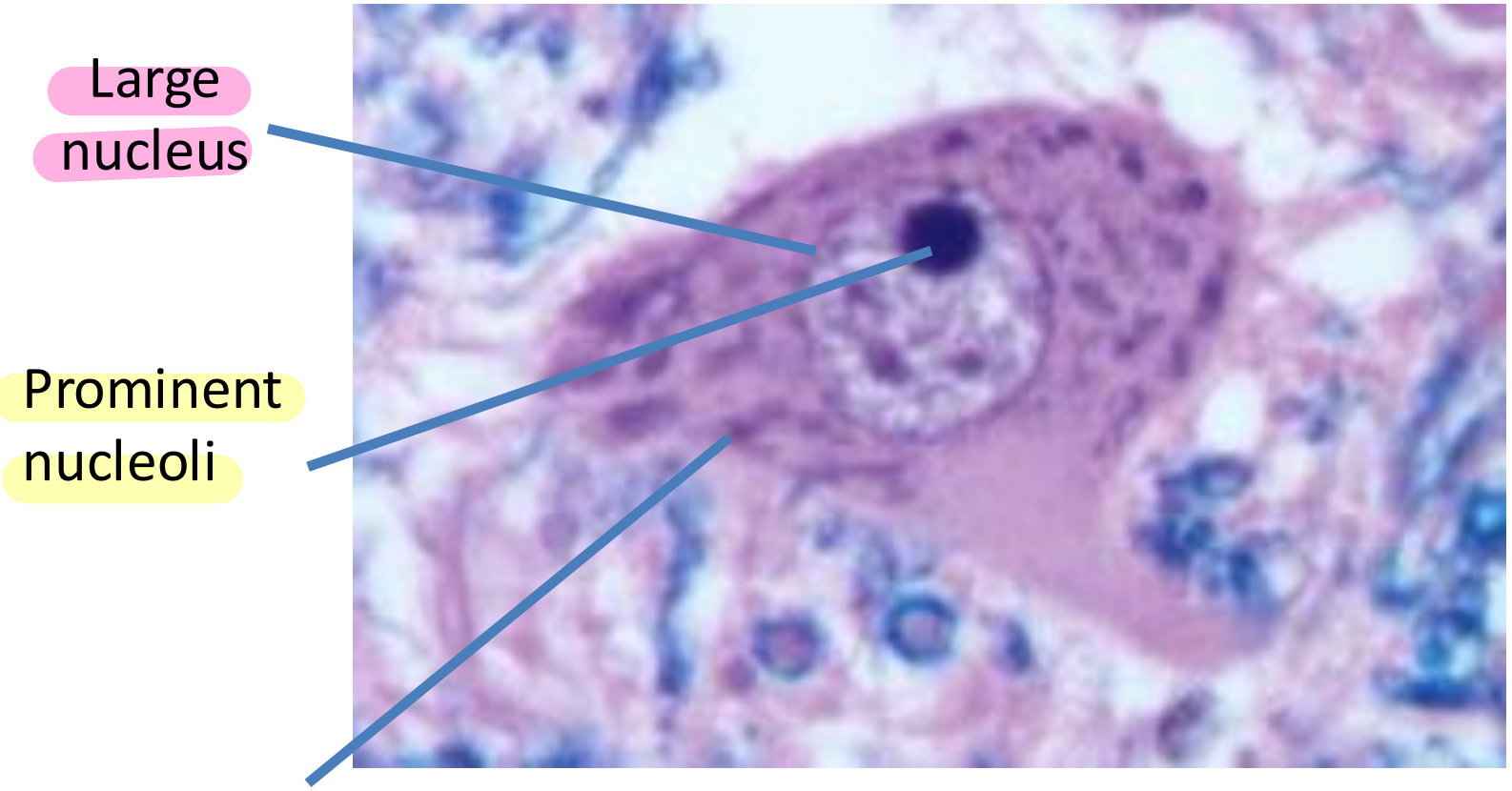


Cell Body “Perikaryon".

- DNA- and RNA containing organelles:
  - Visible on light microscope
- Non DNA- and RNA containing organelles
  - Visible on electron microscope



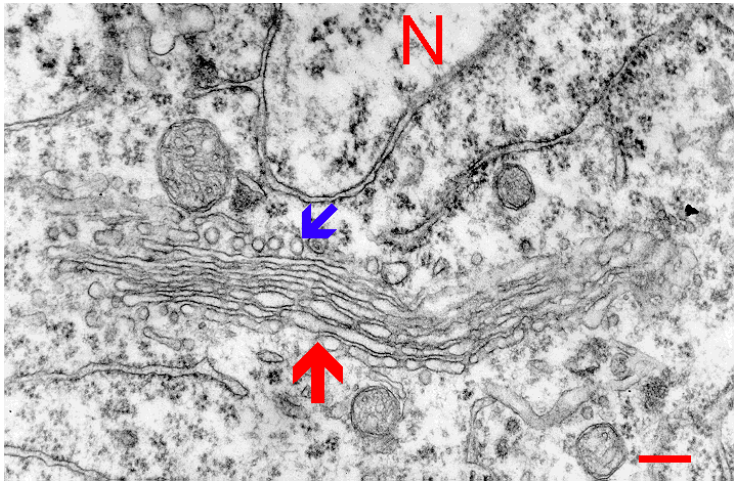
DNA- and RNA Containing Organelles



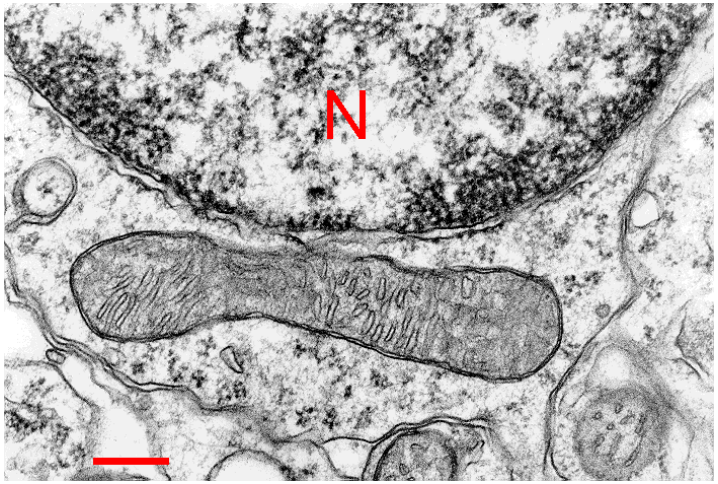
Nissle sunstance are basophilic bodies (abundant cytoplasmic rough endoplasmic reticulum). They are absent in axon and axon hillock.

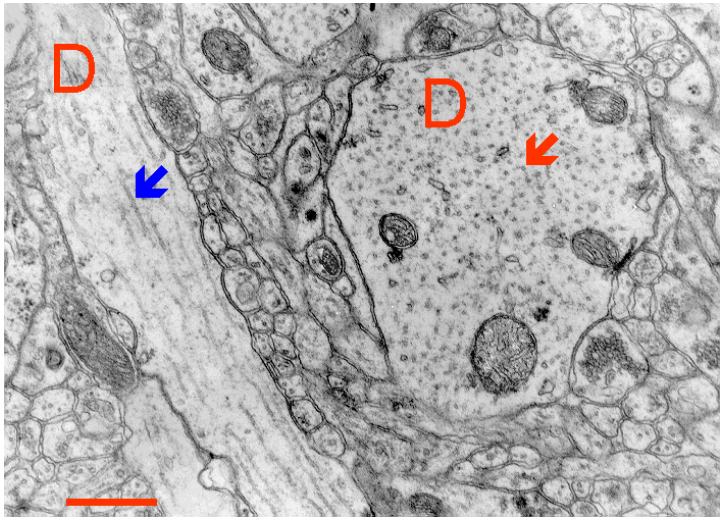
Non DNA- and RNA Containing Organelles

Golgi apparatus

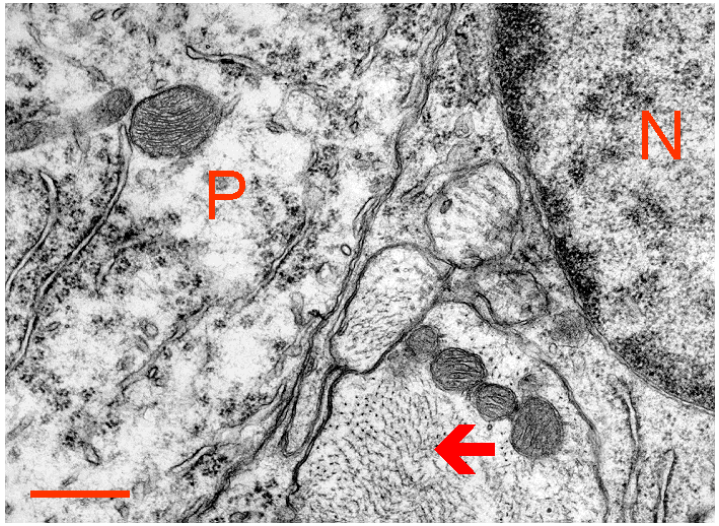


Mitochondrion





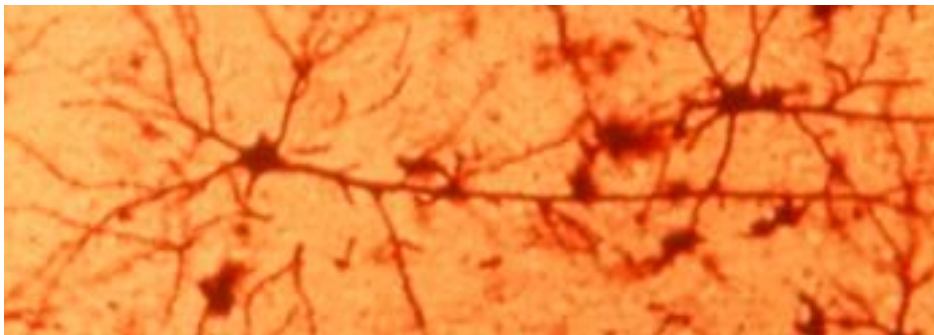
Microtubules



Neurofilaments

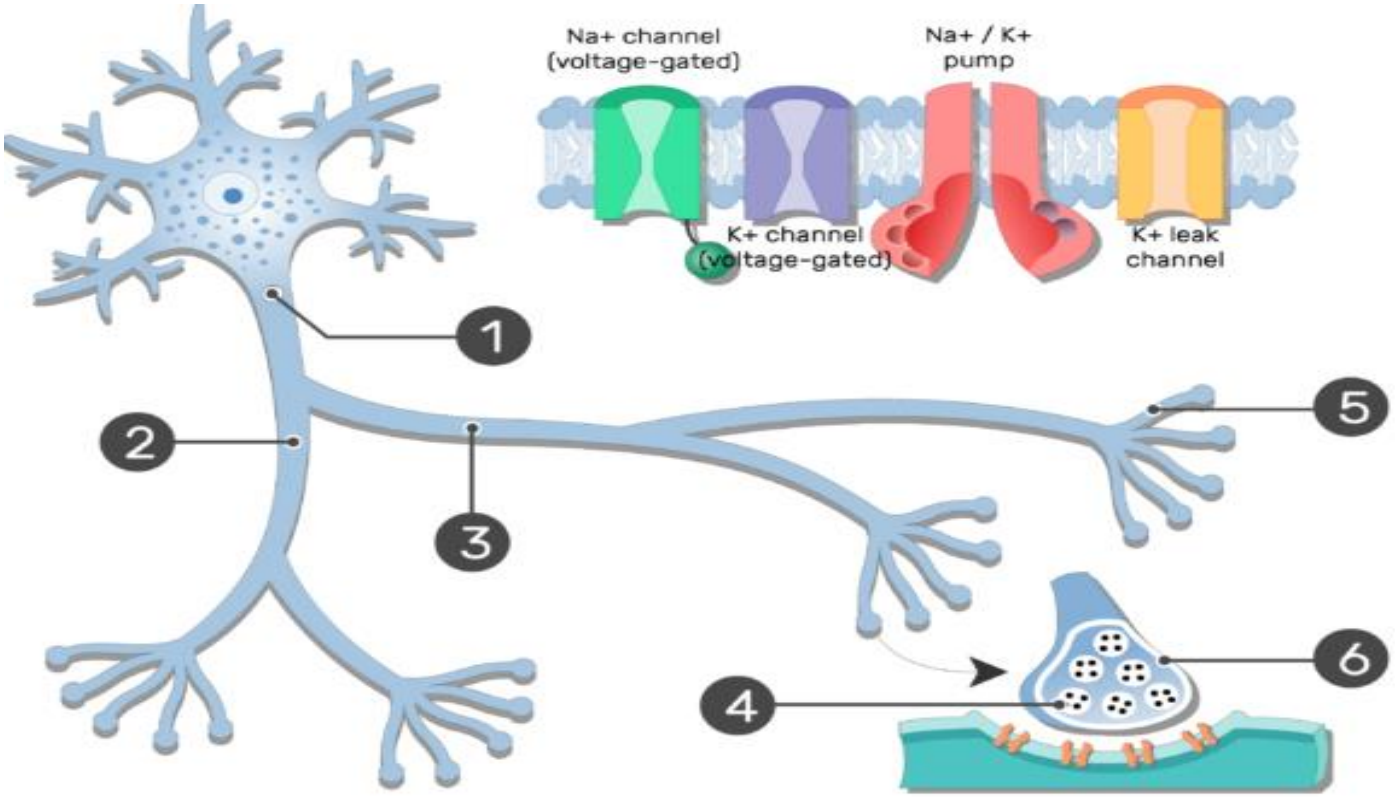
Axon (The Processes of the Neuron)

- Single, thin, elongate process
- Has regular and constant diameter
- Has a smooth surface.
- Branches at its terminal only (terminal arborization).
  - It may give “collaterals"
- Contains mitochondria, neurofibrils and neurotubules.
- No Nissl granules.
- Conduct nerve impulse away from the cell body.
- **May be myelinate**



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Parts of the Axon



1. Axon hillock
2. Axon
3. Collateral branches
4. Synaptic vesicle
5. Terminal branch
6. Synaptic knob

Types of Axons

Type	Myelin	Neurilemma	المكان
Myelinated with Neurilemma	✓	✓	PNS
Myelinated without Neurilemma	✓	×	CNS White Matter
Non-myelinated with Neurilemma	×	✓	PNS، ومنها ألياف Autonomic
Naked Fibers	×	×	CNS Gray Matter

The Nerve Fiber is the axon of the nerve cell.

Axons may be very short (Golgi type II) or very long (Golgi type I).

Naked fibers

Ensheathed fibers

Myelinated with neurilemma: are found in the peripheral nervous system.

Myelinated without neurilemma: are found in the white matter inside the C.N.S.

Non-myelinated with Neurilemma: are found in autonomic nervous system and in peripheral nerves.

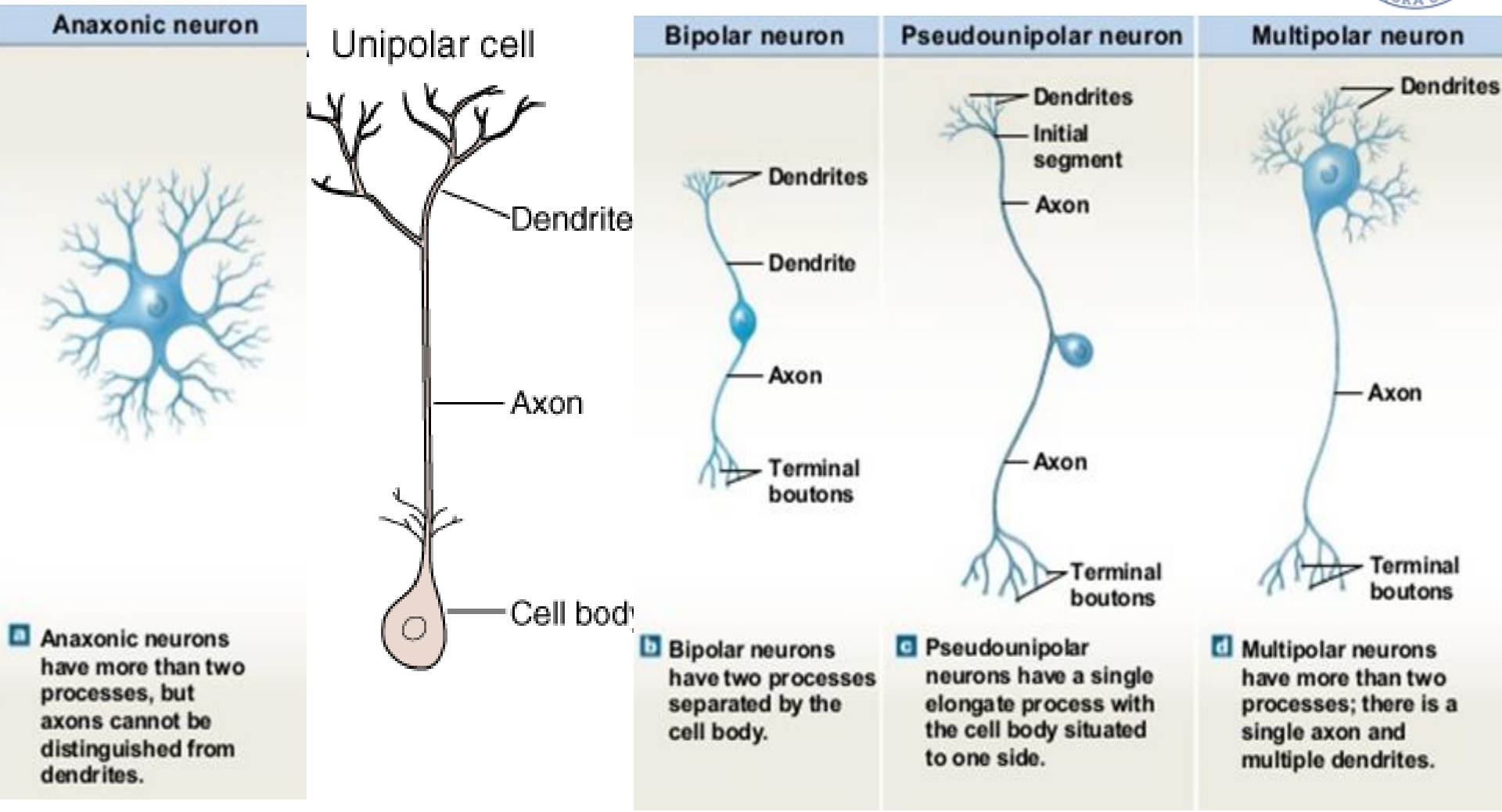
Naked Fibers: Non-myelinated without Neurilemma. They are found in the grey matter of the C.N.S

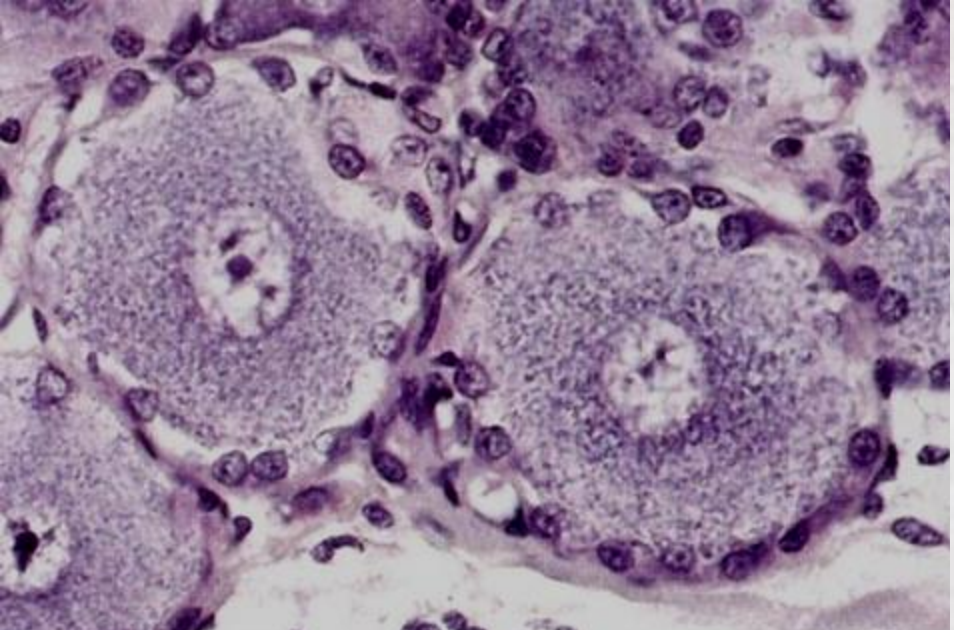
Dendrite (The Processes of the Neuron)

مختلفة عن ال axons

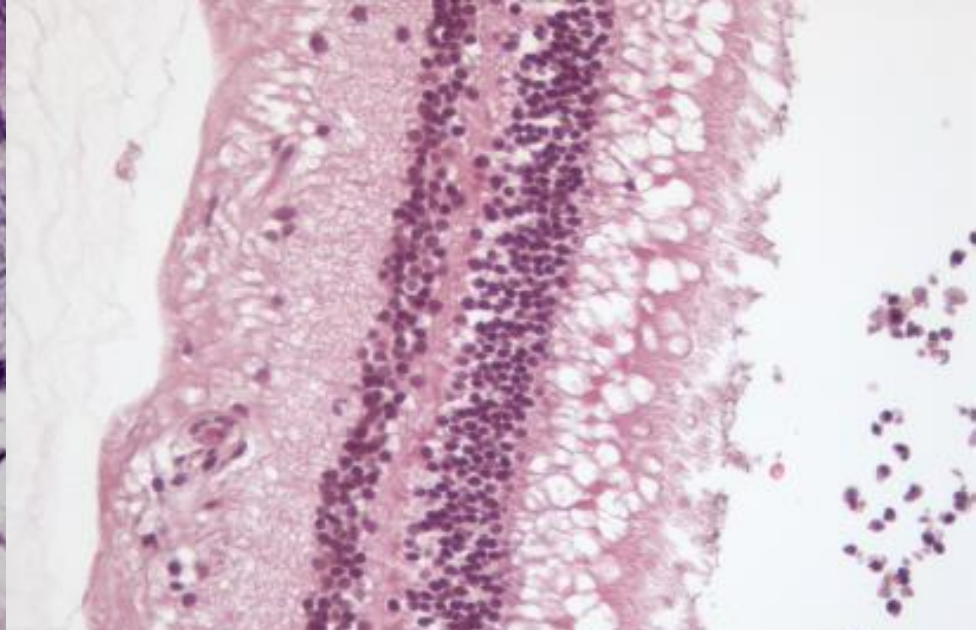
- Usually multiple, thick, and short.
- Thick near its origin, and thin towards its apex.
- Irregular surface which covered by dendritic spines (gemmules).
- Extensive branching along its course the branches arise at acute angles.
- Contain the usual organelles except the Golgi complexes
- Contain Nissl granules
- Conduct nerve impulse to the cell body and the other dendrites.
- **No myelin sheath**

Morphological Classification of Neurons

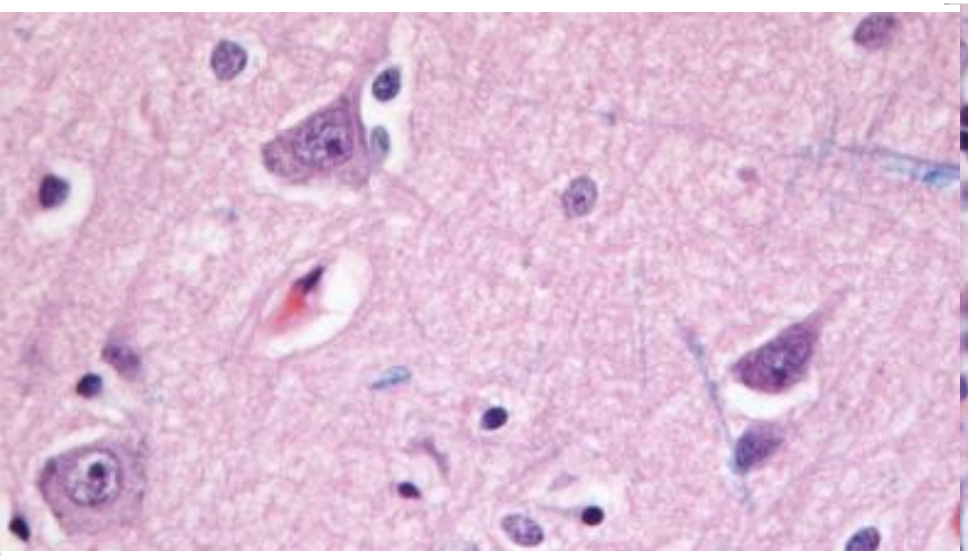




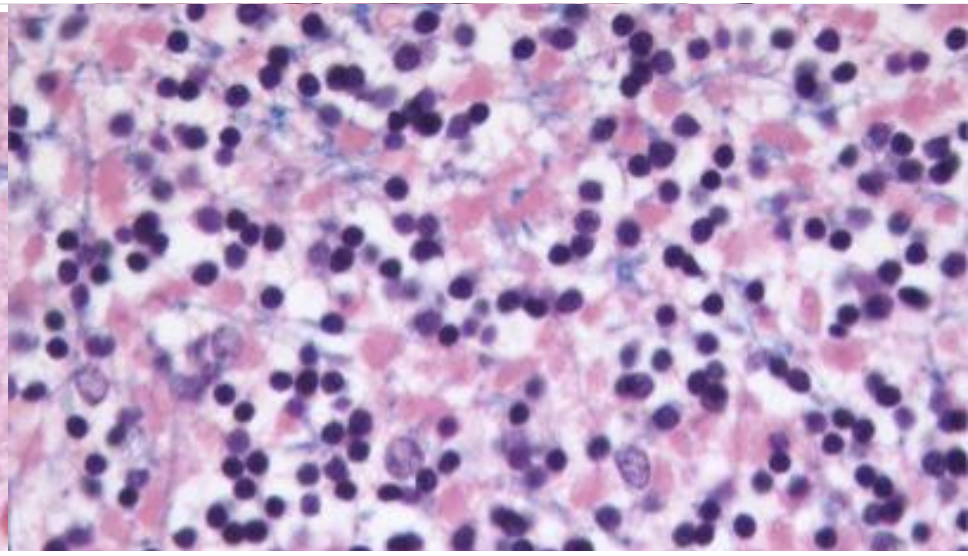
Unipolar, dorsal root ganglia



Retinal bipolar cells



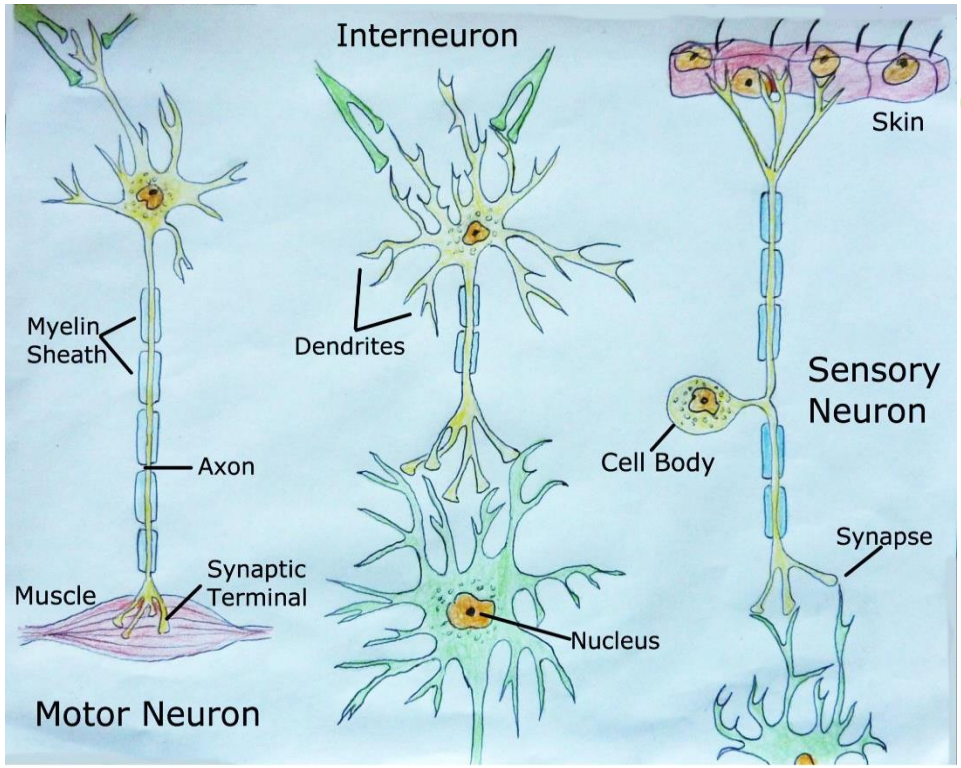
Multipolar: Golgi type I, pyramidal neurons



Multipolar: Golgi type II, Granule neurons of the cerebellum

Functional Classification of Neurons

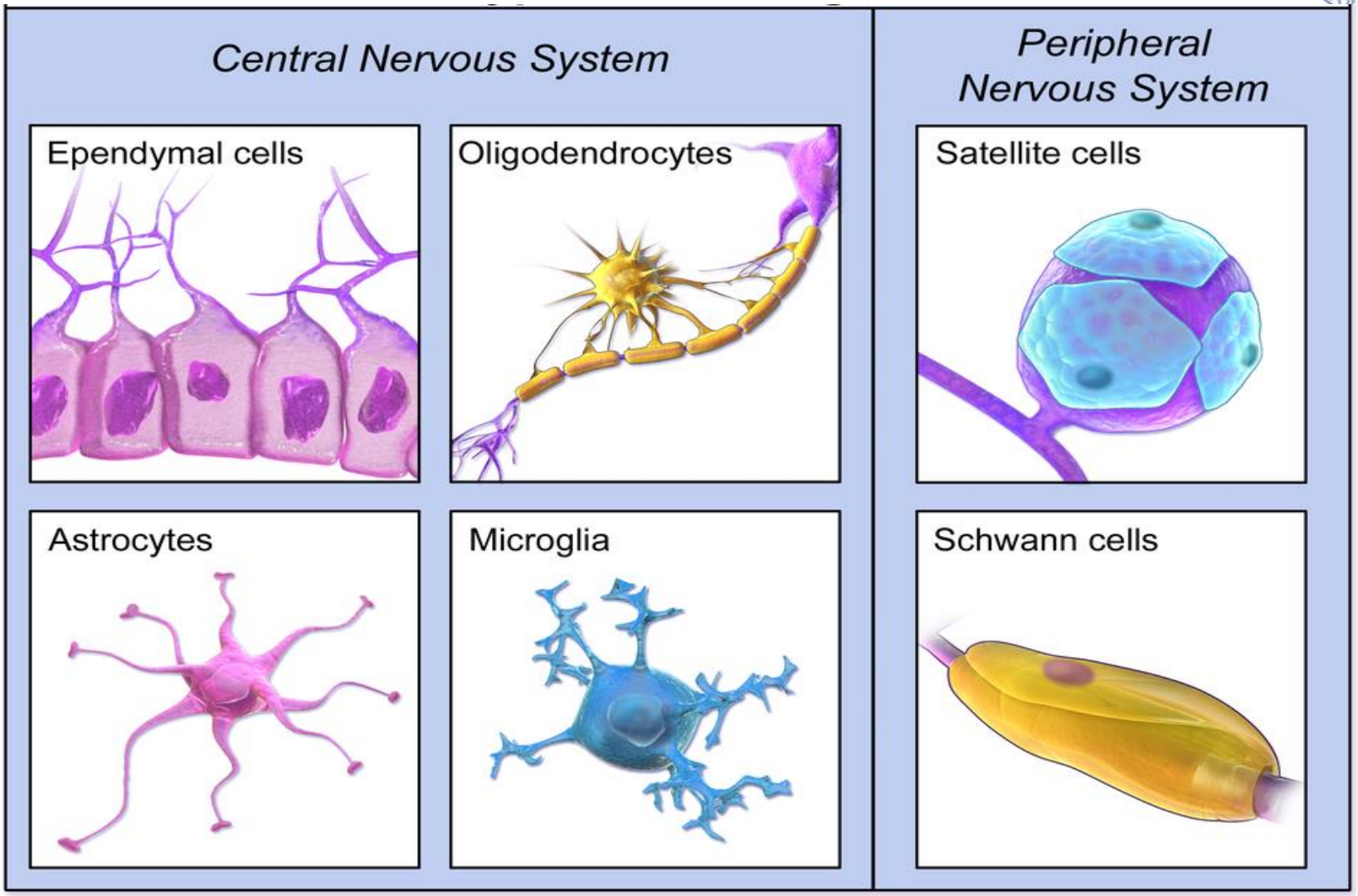
1. Motor neurons conduct impulses from the CNS to other neurons, muscles, and glands



2. Sensory neurons receive stimuli from the internal and external environments

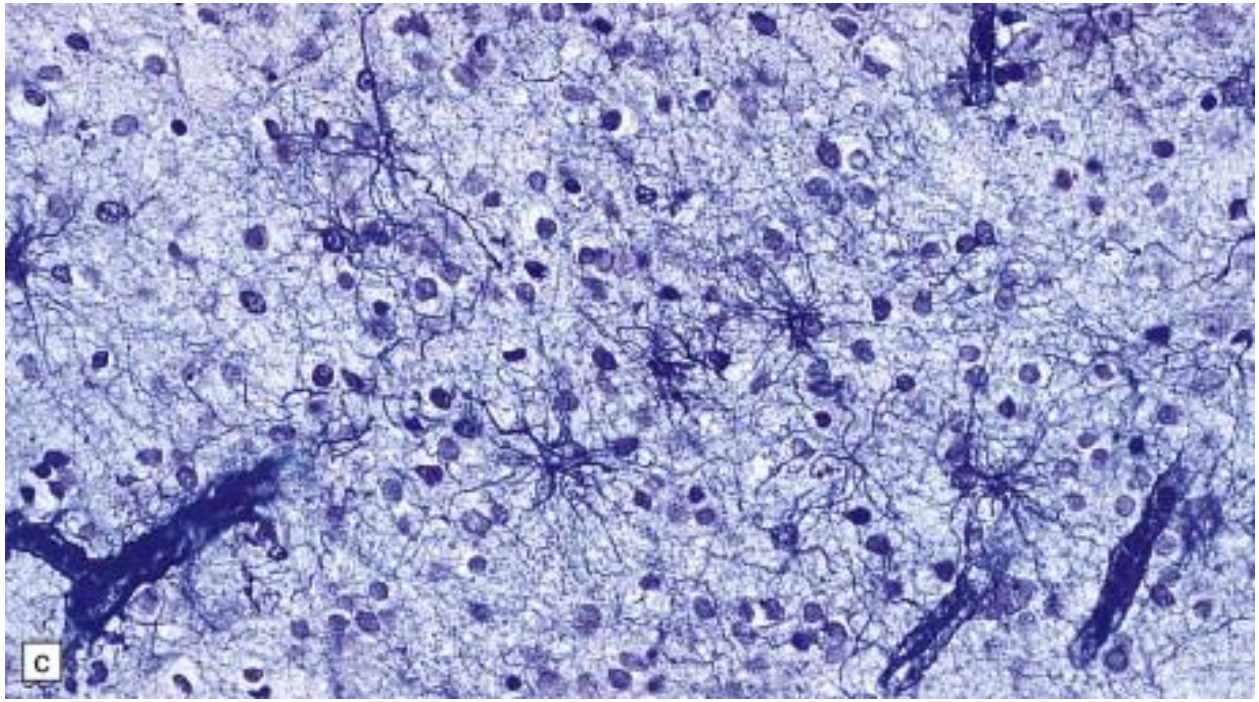
3. interneuron neuronsconnect other neurons in a chain also regulate signals transmitted to neurons

Types of Neuroglia



Astrocytes

- The most abundant type and the largest one
- Owe their name to their irregular, roughly star shaped cell bodies and numerous processes
- Many of their processes end in expansions called end-feet

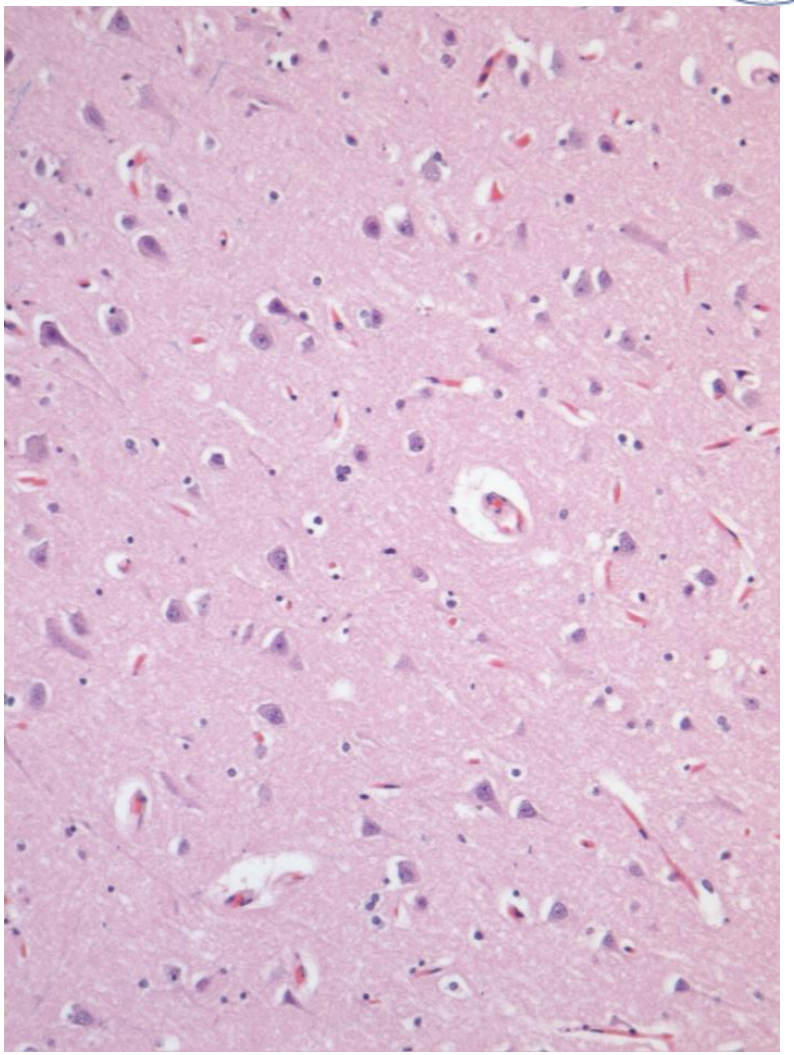


Astrocytes

- Help form blood – brain barrier
- Provide structural support for neurons
- They regulate the external chemical environment of neurons by removing excess K ions .

Astrocytes

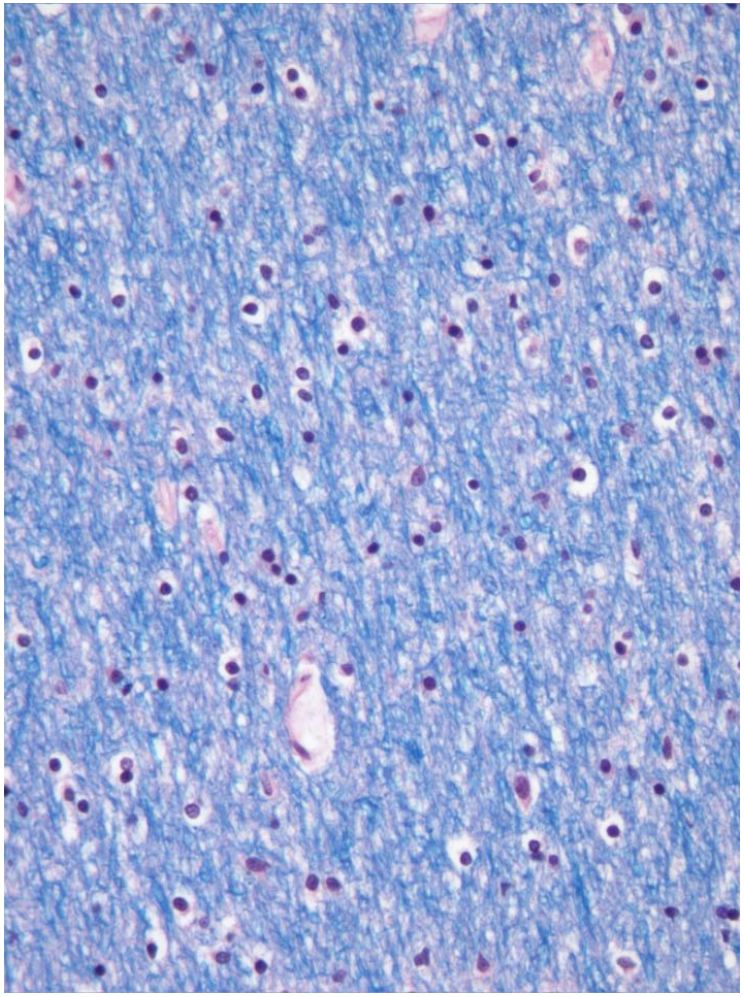
- Nuclei are visible with lighter and looser chromatin than oligodendrocytes
- Nucleoli are not present
- Scant, delicate cytoplasm and processes are not readily seen (blind with the surrounding neuropil)



Oligodendrocytes

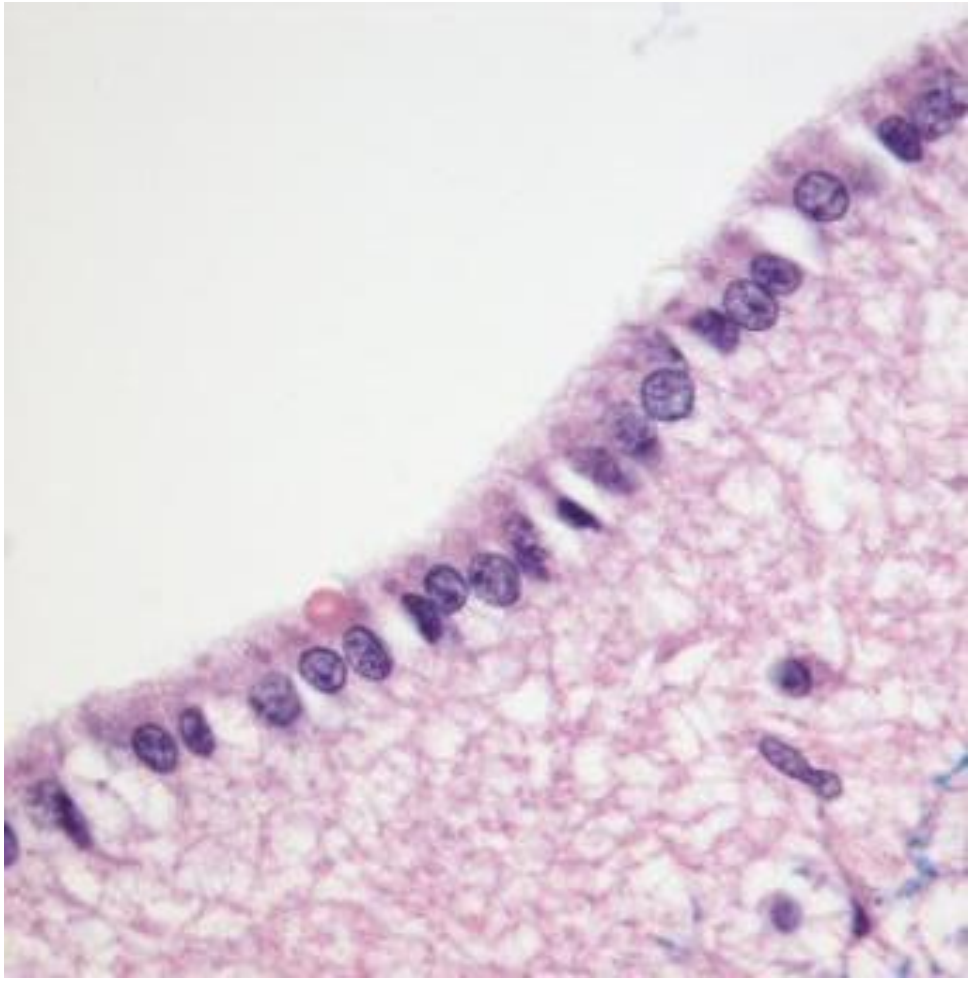
تكون الmyelin

- Greek: oligos, little, few
- Neuroglial cells with small cell bodies and few short cell processes
- In H&E-stained sections:
  - Only the round nucleus is usually visible
- Myelinating cells
- Myelin increases the speed of conduction of impulses

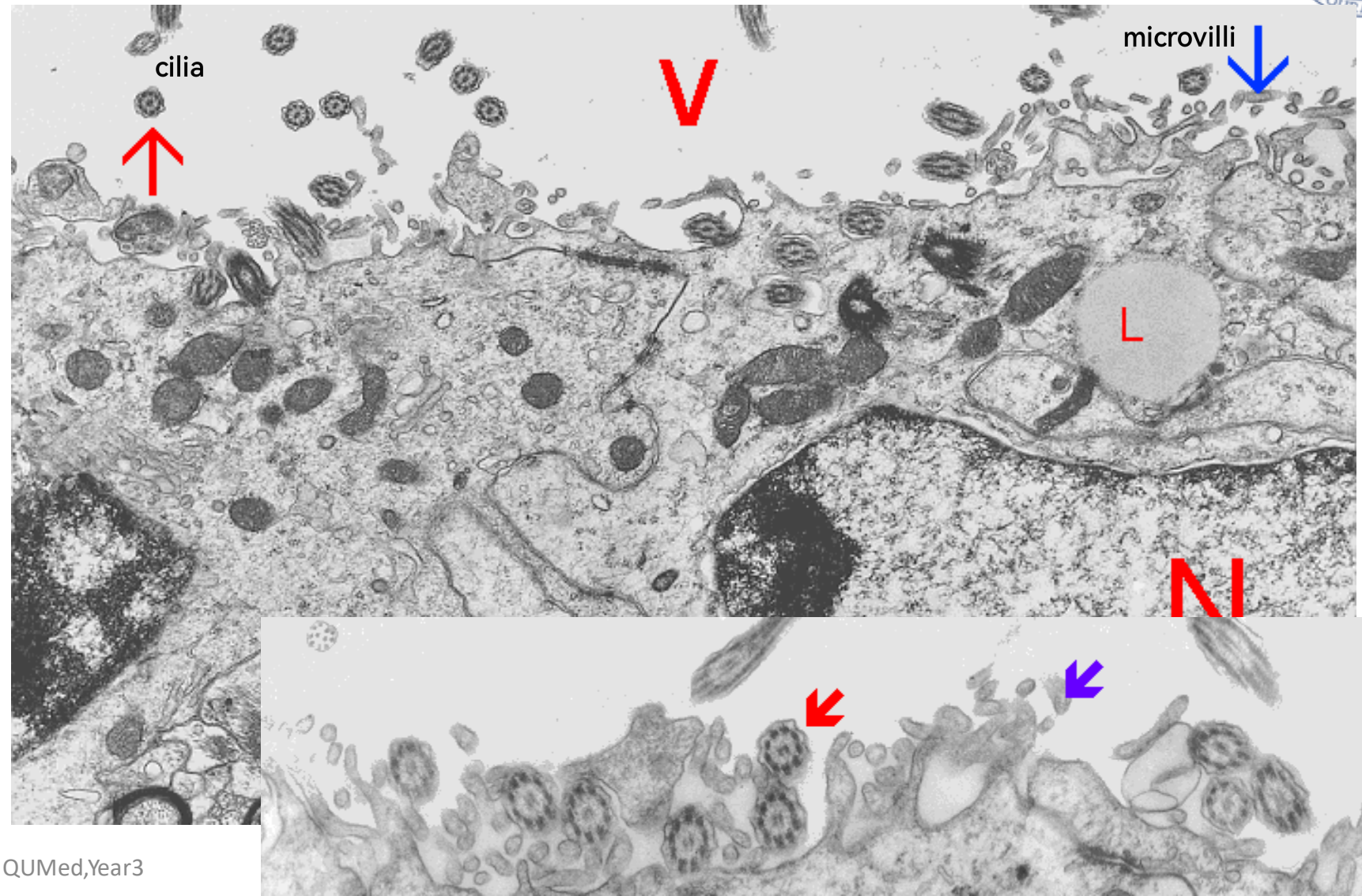


Ependymal cell

- Extending from the lateral ventricles to the filum terminale
- Ependymal cells are cuboidal to culomnar cells
- Round nuclei
- Fine stippled chromatin
- Inconspicuous nucleoli
- Covered by microvilli and most of the cells have a central cluster of long cilia



Ependyma



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Microglia

- Blood-derived monocytes move into the brain during early embryonic development then differentiate into microglia
- Share many surface markers or antigens with their blood-borne (monocytes and macrophages)
- Account for 20% of the cellular population
- Normally not visible on light microscopy
- First and main form of active immune defense in the CNS

Schwann Cells

درسنا سابقًا خلية تصنع الميالين داخل الجهاز العصبي المركزي

Oligodendrocyte

طيب، من يصنع الميالين في الجهاز العصبي الطرفي؟

Schwann Cell

- Schwann are the principal glia of the PNS
- Formation of myelin sheath in the PNS
- There are two main Schwann cells in PNs:

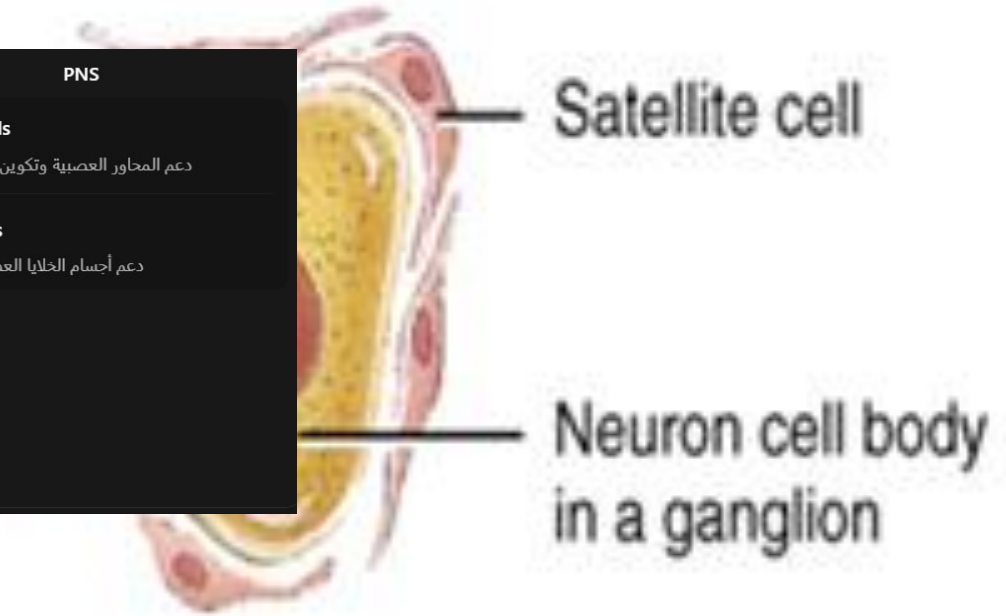
  1– Myelinating schwann cells which surround large axons

  2– Non-myelinating schwann cells which enclose smaller axons in invaginations of their surfaces

Satellite Cell

- Glial cells that cover the surface of nerve cell bodies in sensory, sympathetic and parasympathetic ganglia
- Derived from the neural crest

CNS	PNS
Astrocytes تنظيم البيئة والدعم	**Schwann Cells** دعم المحاور العصبية وتكوين الميالين لبعضها
Oligodendrocytes تصنيع الميالين	**Satellite Cells** دعم أجسام الخلايا العصبية داخل العقد
Microglia الدفاع المناعي	
Ependymal Cells تبطين تجاويف السائل	

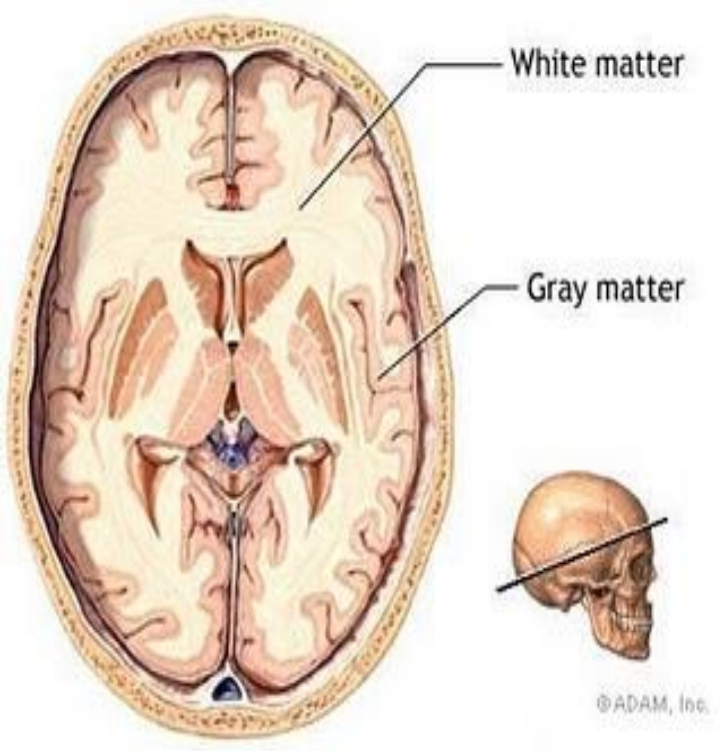


Cytoarchitectural Organization of the CNS

- Cerebrum
- Cerebellum
- Brain stem
- Spinal cord

Cerebrum

- Grey matter:
  - Cortex
  - Deep grey structures
- White matter

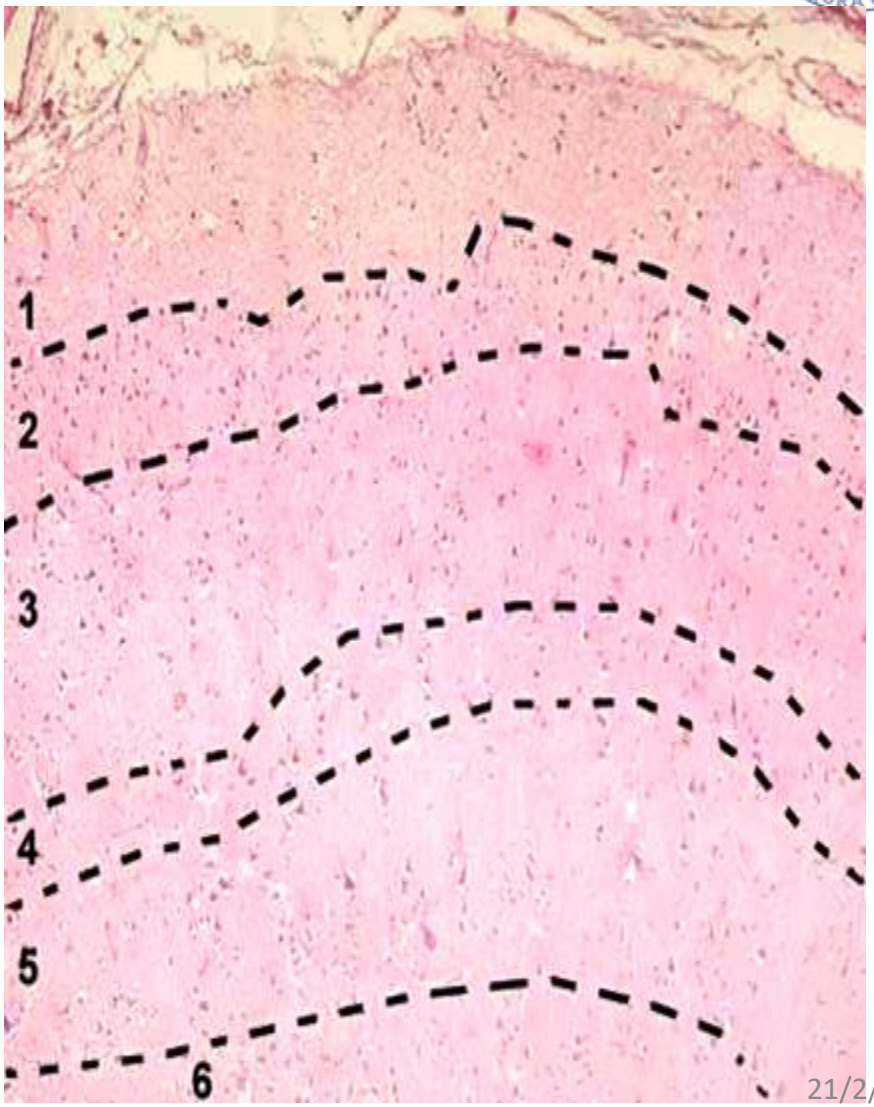


Cerebral Cortex

- Superficial gray matter
- Classification
  - Based on the differences in lamination and development
    1. Isocortex (homotypical cortex), part of the cortex with 6 layers, 90% of the cerebral cortex
    2. Allocortex (heterotypical cortex) the part of the cortex with less than 6 layers (varying in number), 10% of the cerebral cortex

Isocortex/ layers

- 6 layers:
  1. Molecular layer
  2. Outer granular layer
  3. Outer pyramidal layer
  4. Inner granular layer
  5. Inner pyramidal layer
  6. Multiform cell layer



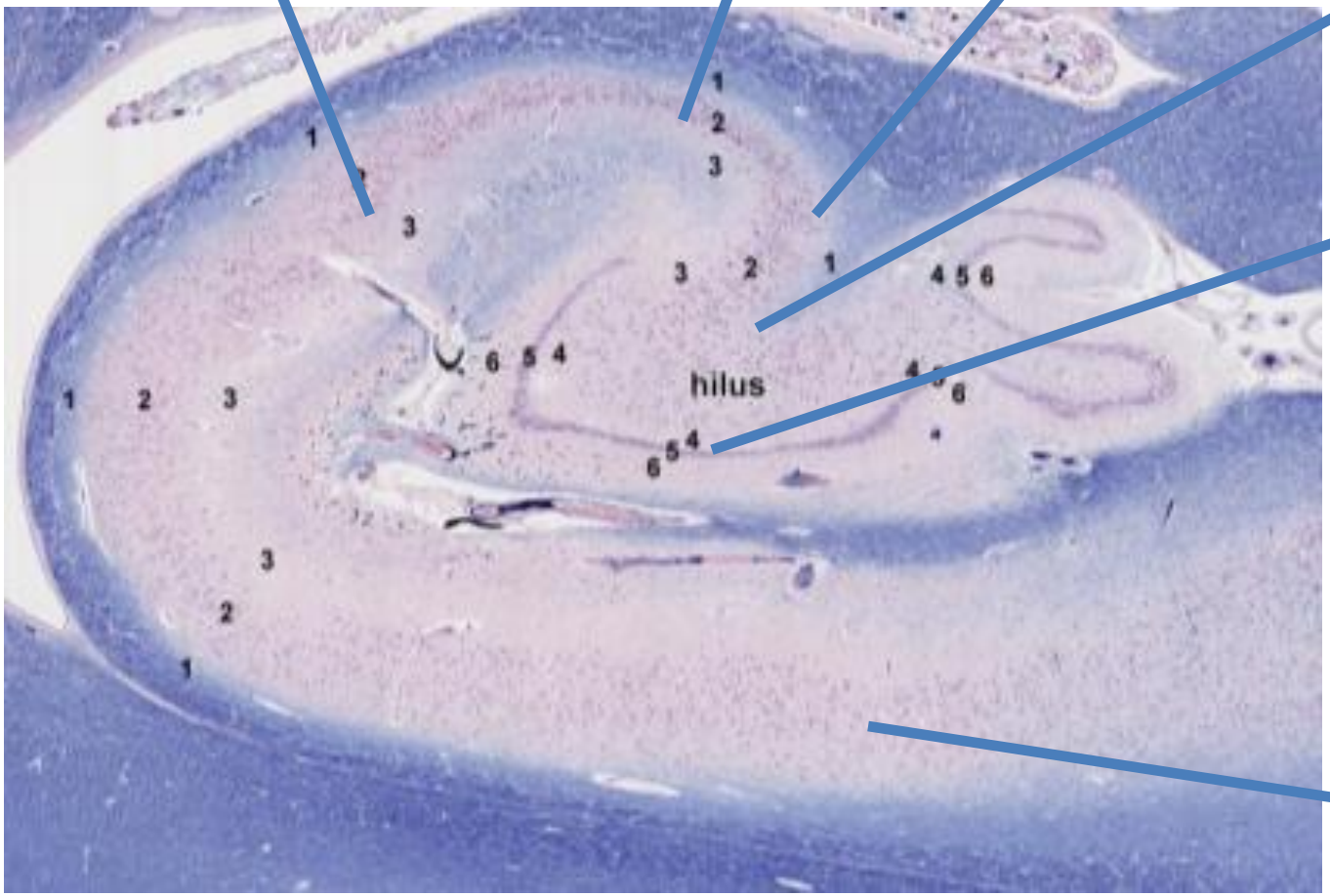
Allocortex

- Heterotypical cortex
- The part of the cortex with less than 6 layers (varying in number)
- Types:
  - Archicortex, hippocampal structure
  - Paleocortex, mostly limbic and olfactory cortices

Archicortex (Hippocampal Formation)

- Hippocampus proper
- Dentate gyrus
- Subiculum



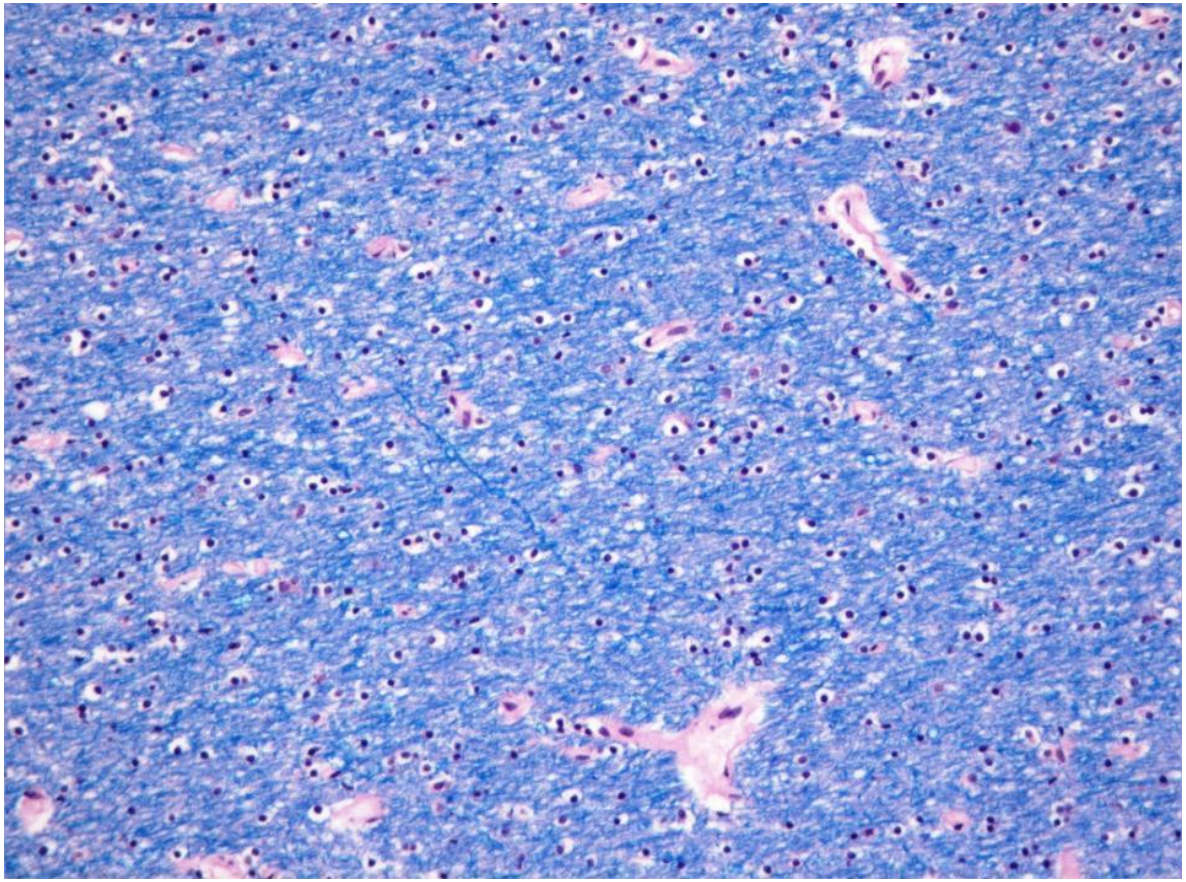

CA1
CA2
CA3
CA4
Dentate gyrus
hilus
Subiculum

Paleocortex

- A layer of the cerebral cortex intermediate between the neocortex and archicortex
- Entorhinal cortex:
  - Occupies most of the parahippocampal gyrus
  - Largest source of input into the hippocampus

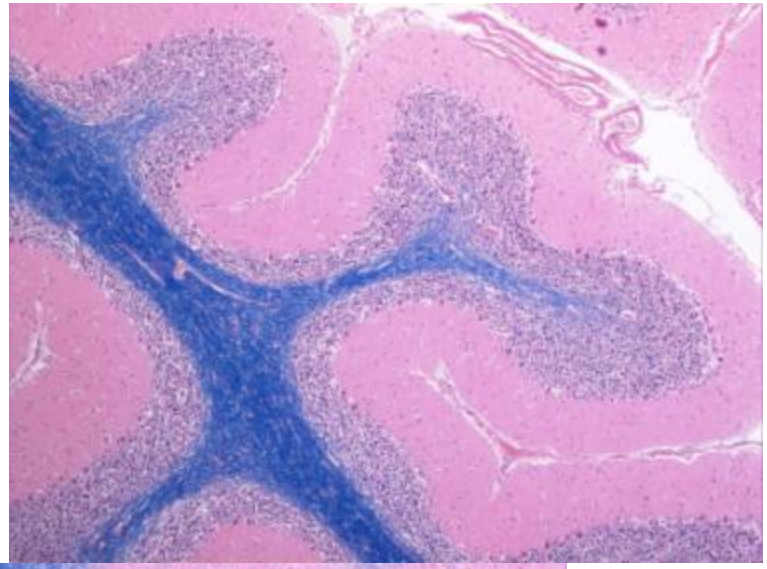
White Matter

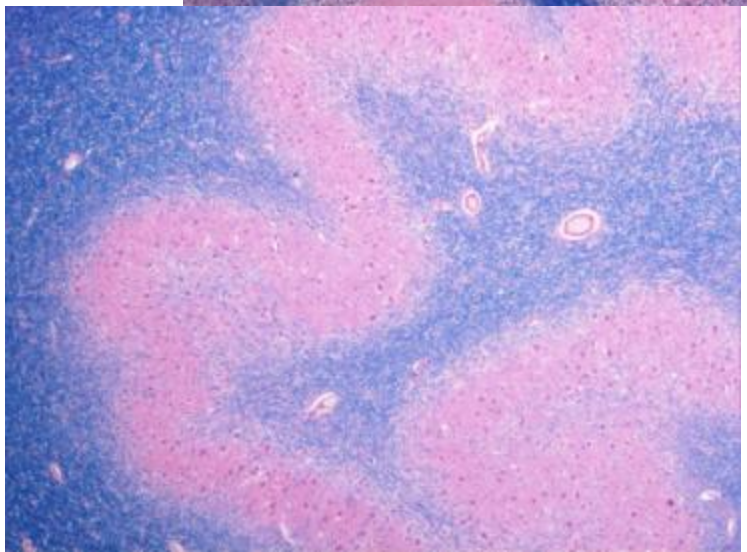
- Myelinated axons and oligodendrocytes خلايا تصنع myalien



Cerebellum

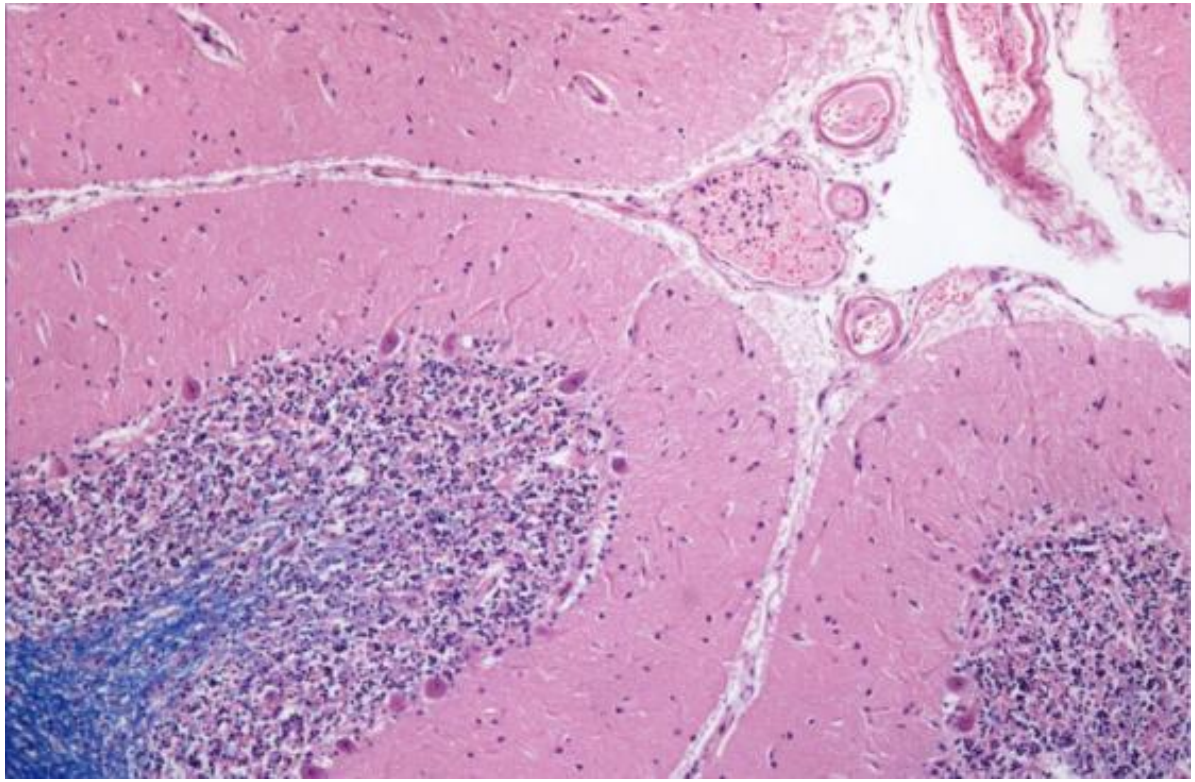
- Cortex
  - Three layers
- White matter
  - Myelinated axons
  - Deep cerebellar nuclei





Cerebellar Cortex

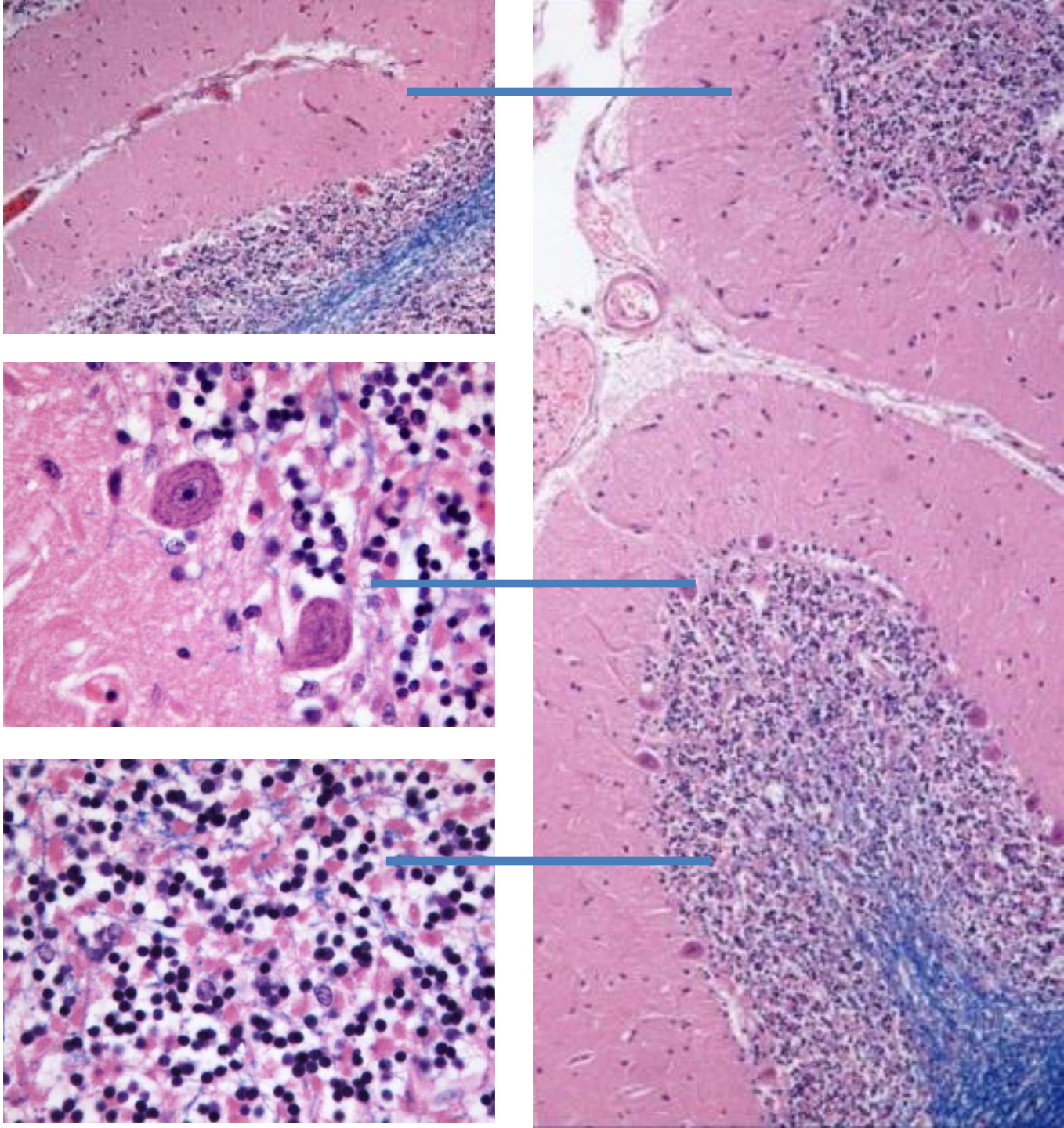
1. Molecular layer
2. Purkinje cells
3. Granular layer



1. Molecular layer

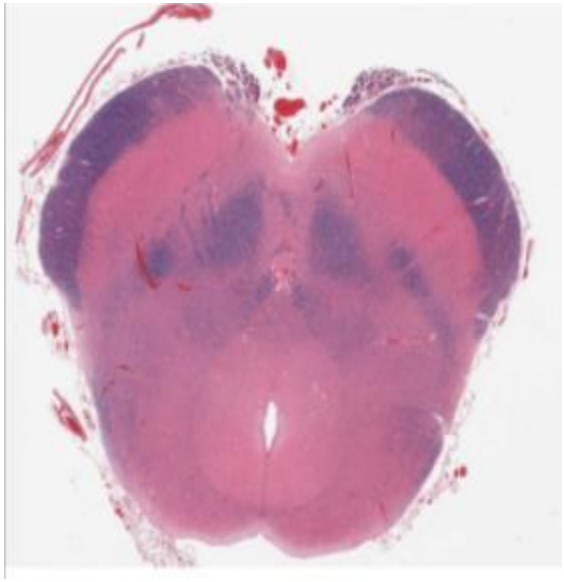
2. Purkinje cells

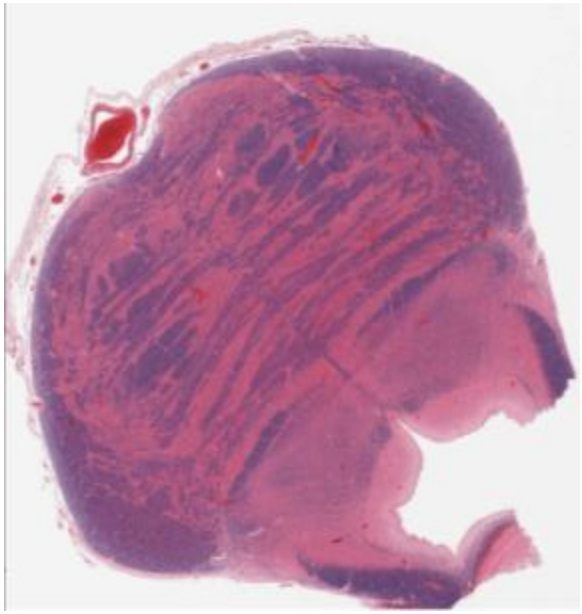
3. Granular layer

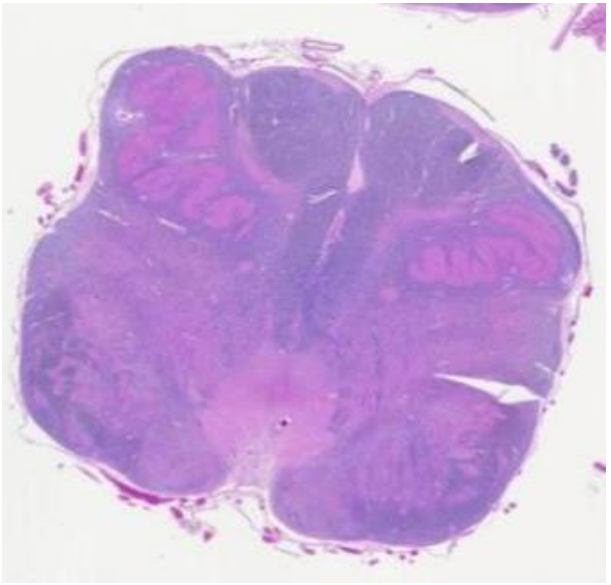


Brain Stem

- Midbrain
- Pons
- Medulla oblongata

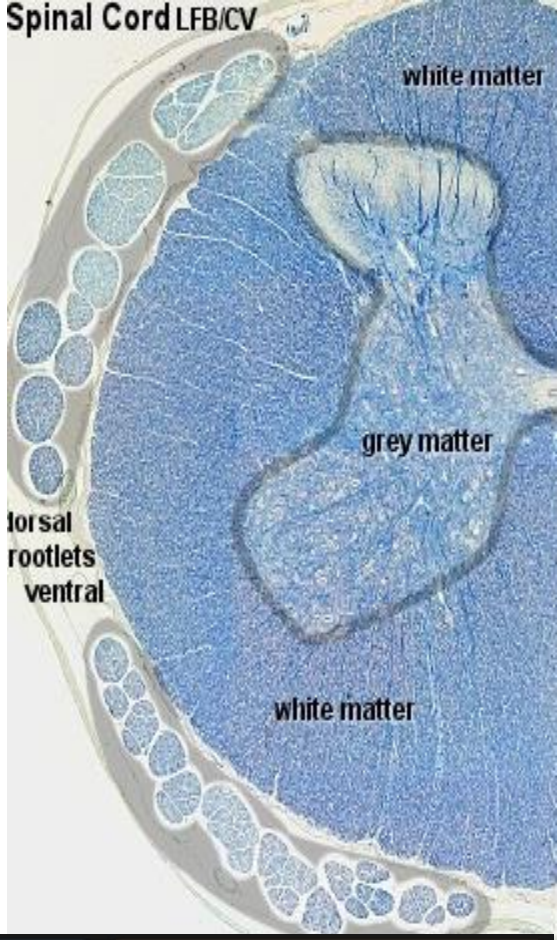


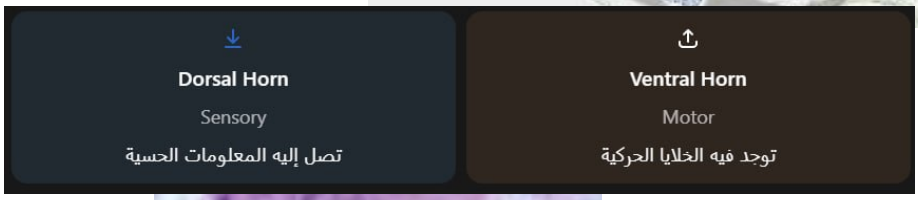




Spinal Cord

- A small central canal, lined by ependymal cells, This canal is a remnant of the embryonic neural tube.
- Dorsal horns consist of sensory fibers extending from the dorsal root ganglia.
- Ventral horns consist of cell bodies and fibers of large multipolar motor neurons.



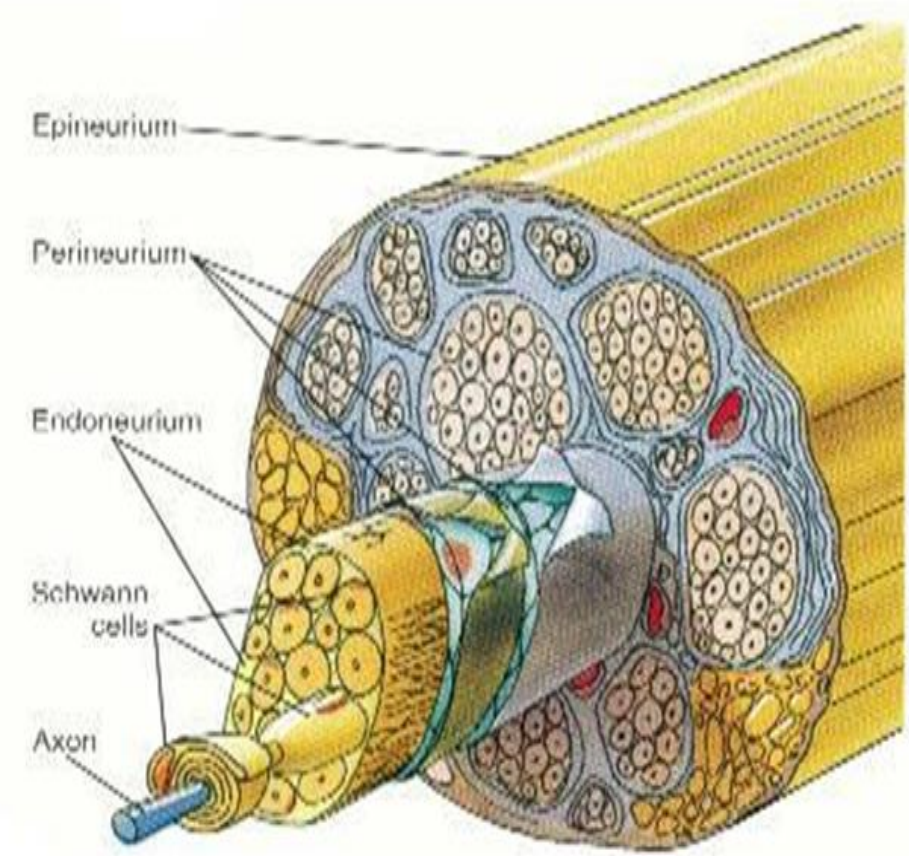




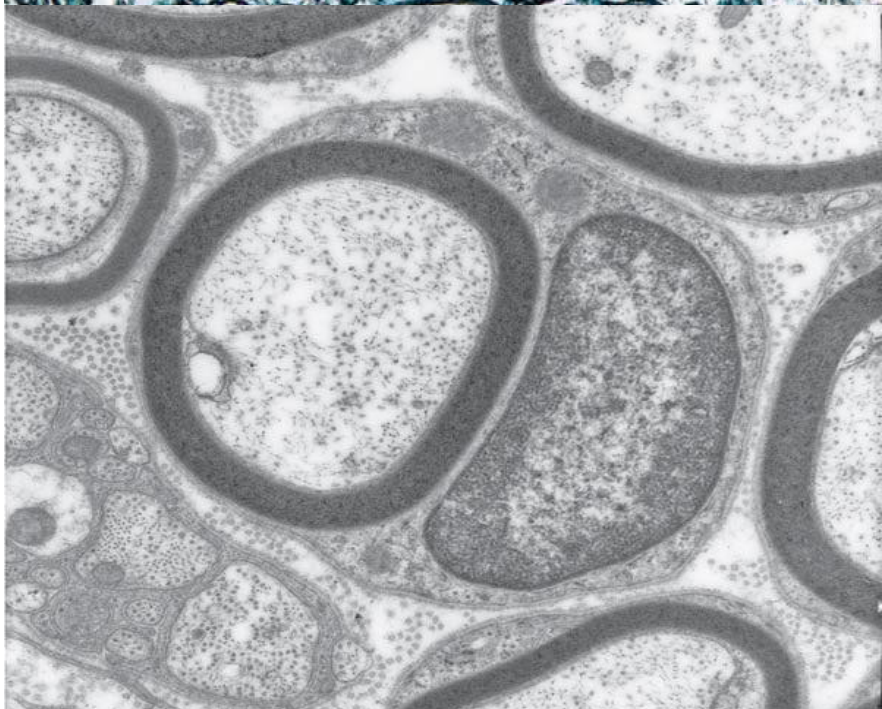
Peripheral Nerve

- The peripheral nerve consists of several nerve fibers which are grouped in bundles of different sizes
- May be motor or sensory or mixed.
- Consisting of bundles (Fascicles) of nerve fibres.
- Supporting connective tissue makes layers:
  - Endoneurium: around each nerve fire. ليفة عضلية على حدى
  - Perineurium: around each fascicle.
  - Epineurium: around the whole nerve.

Structures of a peripheral nerve trunk



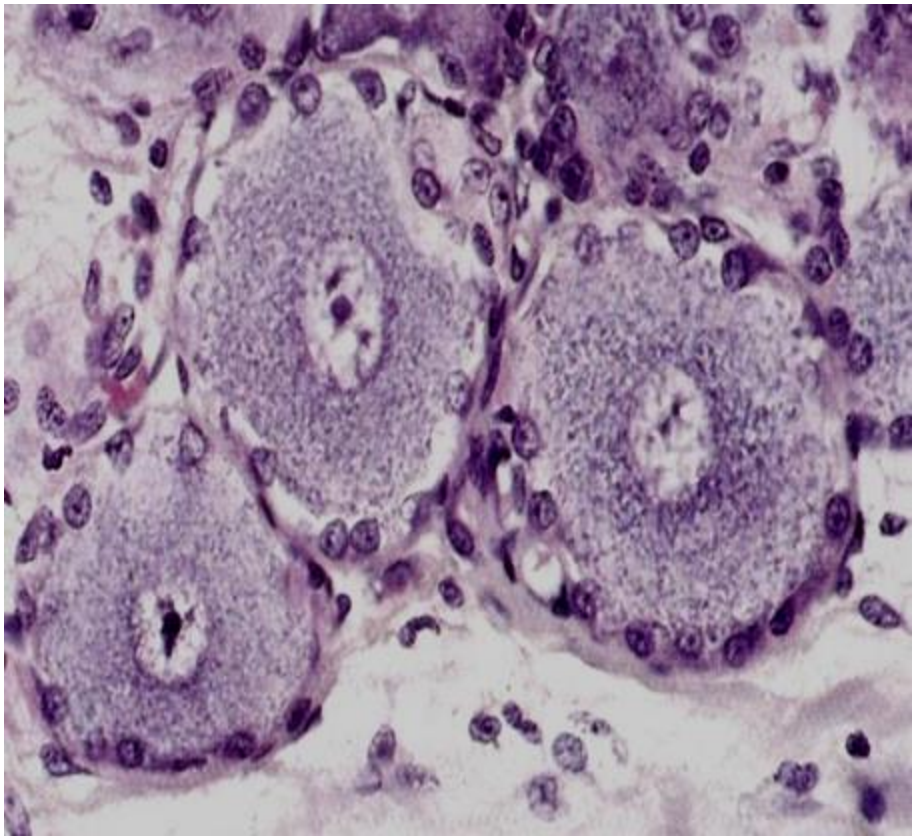




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Ganglia

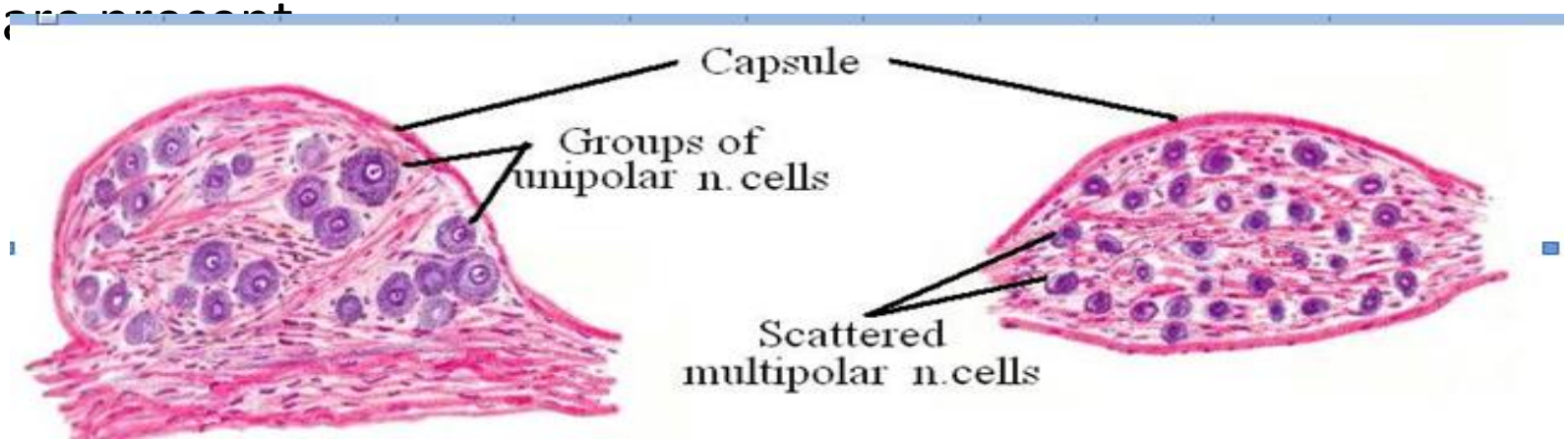
- Spinal ganglion:
  - A sensory ganglion located on spinal nerves.
  - Dorsal root ganglion



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Ganglia

- Autonomic ganglia
  - Sympathetic chain and parasympathetic nerves
- The capsule is thin or absent and the cells are supported by the stroma
- Neurons are multipolar, with eccentric rounded nucleus and scattered in the gangila - surrounded by few satellite cells.
- The preganglionic nerve fibers are unmyelinated and synapses are present



Spinal	Autonomic
Capsule	
thick	thin
Nerve cell:	
Pseudounipolar	multipolar
Rounded cell body	Stellate
Arranged in groups	not arranged in groups
large in size and few	Small and many
Rounded central nucleus	Eccentric nucleus
Surrounded by satellite (Capsular)cells	Few satellite cells
Presence of glomerlus	No glomerlus
Nerve fibers:	
mylinated	Smaller not mylinated
Site:	
cranial and spinal sensory nerves	Sympathetic chain and along the parasympathetic nerves.

	CNS	PNS
Grey matter: Nerve cell collections	**Cortex** (on surface) **Nuclei** (deep)	Ganglion
White matter Nerve fibres	Tracts	Nerves

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Questions

- Nissl bodies are composed of

1. Synaptic vesicles and acetylcholine.
2. Polyribosomes and rough endoplasmic reticulum.
3. Lipoprotein and melanin.
4. Neurofilaments and microtubules.
5. Smooth endoplasmic reticulum and mitochondria. only.



References

- **Gartner, L. P. and Hiatt, J. L. (2007):** Color textbook of histology, 3rd edition, W. B. Saunders Company, Philadelphia, London, Toronto, Montreal, Sydney and Tokyo, chapter 11, pages 165:172.
- **Gartner, L. P.; Hiatt, J. L. and Strum, J. M. (2016):** Cell biology and histology, 6th edition, Wolters Kluwer, Health and Lippincott Williams and Wilkins. Philadelphia. Baltimore. NewYork. London etc.ch 12,p 180:193.